Reproducing Embedded Wi-Fi Sensing for Human Activity Recognition: A Replication Study on ESP32 Hardware

Swastik Kumar, Deepak Kumar, Satvik Rastogi, Vansh Sharma, and Neeru Jindal

***Abstract*—Wi-Fi channel state information (CSI) collected on low-cost microcontrollers is an attractive basis for privacy-preserving human activity recognition (HAR). A recent study reported a complete embedded pipeline—CSI capture on ESP32 devices, empirical mode decomposition (EMD) augmentation, and an LSTM-embedded DenseNet running on an ESP32-S3—with 92.43% accuracy in 127 kB. We reproduce this work from its public release, which provides data, collection firmware and trained models but no training, inference or deployment code. We rebuild the model weight-exactly, recover the held-out subjects, train with the published hyperparameters for the full 300 epochs, and rebuild the two-board CSI collection system. Under the original protocol our model reaches 94.56% per-subcarrier and 97.10% majority-vote accuracy in floating point, exceeding the released model, and 90.87% when quantized to 8 bits for the single-subcarrier on-device path. The replication also surfaces several undisclosed details that materially affect the reported numbers: synthetic training samples are built from the test subjects' recordings; the model is trained and evaluated per subcarrier; five specific subcarriers are unusable; the headline 8-bit figure equals the released model's floating-point score; and the released quantized model occupies 322 kB rather than 127 kB. In a new room, the model falls to near chance (26.8% per subcarrier) and is not recovered by fine-tuning or by the original augmentation; a model-free motion-energy analysis shows that most of the signal variation there is not caused by the subject. We release the reconstructed pipeline, host-verified embedded preprocessing and a PlatformIO build of the collection firmware to support a leak-free re-evaluation.**

***Index Terms*—Channel state information, ESP32, human activity recognition, reproducibility, TinyML, Wi-Fi sensing, data leakage.**

I. Introduction

Wi-Fi sensing infers human presence and motion from the way a body perturbs radio propagation between a transmitter and a receiver [1], and has been used to recognize everyday activities [2]. The fine-grained quantity used for this is channel state information (CSI): the per-subcarrier complex channel estimate that an OFDM receiver computes for every packet. Early CSI work depended on specific network cards with modified firmware [3]; the ESP32 family later made CSI available on microcontrollers costing a few dollars [4], opening the way to sensing systems that capture *and* classify on the device, with no camera and no cloud.

The authors are with the Department of Electronics and Communication Engineering, Thapar Institute of Engineering & Technology, Patiala, Punjab 147004, India. N. Jindal is an Assistant Professor in the department (e-mail: skumar6_be24@thapar.edu; dkumar1_be24@thapar.edu; srastogi_be24@thapar.edu; vsharma4_be24@thapar.edu; neeru.jindal@thapar.edu). (Corresponding author: Neeru Jindal.)

Armenta-Garcia *et al.* [5] presented such a system: a configurable CSI collection tool for a pair of ESP32 boards, a 22-subject dataset of five activities, EMD-based data augmentation, and a reduced LSTM-embedded DenseNet [6], [7] deployed with TensorFlow Lite Micro [8] on an ESP32-S3. They reported 92.43% accuracy at 232 ms per inference in 127 kB of memory, and released the data, collection firmware and trained models under an open licence [9].

This paper is a replication of that work. Our goal was practical: to stand up the same pipeline on our own hardware as a basis for further research. Doing so required reconstructing every component that was not released—training, preprocessing for inference, quantization, and on-device inference—and, in the process, exposed details of the original evaluation that are absent from the paper but change how its numbers should be read. Data leakage of this kind is a known source of irreproducible results in machine-learning-based science [10].

Our contributions are:

1) A complete, open reconstruction of the pipeline, with the model architecture verified *weight-exactly* against the released checkpoints and the held-out subjects recovered exactly from the released training array.
2) A full-budget reproduction under the original protocol (300 epochs, published hyperparameters) that matches or exceeds the released model on all scorings.
3) Six findings about the original evaluation and artefacts (Section IV), including test-subject leakage through augmentation and the identity of the unnamed "problematic" subcarriers.
4) A working reproduction of the two-board ESP32 CSI collection system, with the configuration and robustness fixes the released firmware needs, and embedded preprocessing verified bit-exactly against the host implementation.
5) A cross-environment test in a new room, with fine-tuning, augmentation, gain normalization and a fit-check diagnostic, showing that the original pipeline does not transfer and why.

II. The Original System

A. Data Collection

Two ESP32 boards act as transmitter (Tx) and receiver (Rx) on a 20 MHz channel. The Rx operates as an access point; the Tx joins it and sends UDP packets at a configured rate, and the Rx reports the CSI estimated for each packet from the

TABLE I
RESULTS REPORTED IN [5]

Model	Float	8-bit	Latency	Memory
LSTM-Emb. DenseNet	97.33%	92.43%	232 ms	127 kB
LSTM-DenseNet121	97.55%	95.55%	817 ms	277 kB

8-bit figures: one subcarrier. All 47 subcarriers: 93.32%, 10,904 ms.

Tx's MAC address. With legacy (non-HT) packets the CSI buffer is 128 bytes: 64 subcarriers as interleaved signed 8-bit imaginary/real pairs. Data leave the Rx over USB serial, GPIO UART or an SD card, in ASCII or binary form; the authors report an empirical ceiling of about 85 packets/s for binary output over USB.

B. Dataset

Twenty-two volunteers performed five activities—fall (FA), get up (GU), lie down (LD), sit down (SD) and walk (WA)—about 15 times each for 20 s at 50 packets/s in a computer laboratory, giving 1,647 recordings. The released raw files contain between 866 and 1,001 packets each (median 988).

C. Preprocessing and Augmentation

CSI amplitude is computed per subcarrier, the DC null and guard-band subcarriers are removed (52 remain), and each recording is truncated to its first 850 packets. The paper states that 47 subcarriers were finally used and that a single subcarrier with the highest variance is selected for deployment. For augmentation, ensemble EMD [11], [12] with 100 trials decomposes each subcarrier into seven intrinsic mode functions plus a residue; a synthetic sample sums the even-indexed modes of one recording with the odd-indexed modes of another recording of the same subject and activity. The paper reports 11,513 synthetic samples from 1,198 real training recordings.

D. Model and Deployment

The "LSTM-embedded DenseNet" replaces DenseNet's 2-D convolutions with 1-D ones, uses four dense blocks of 3, 6, 12 and 8 layers with growth rate 11, and inserts a 32-unit LSTM before global pooling (122,383 parameters). A larger variant with DenseNet-121 block sizes and a 128-unit LSTM has 437,234 parameters. Training uses Adam at 10^{-4}, batch size 32 and 300 epochs. The models are quantized to 8 bits and run under TensorFlow Lite Micro on an ESP32-S3 with external PSRAM. Table I lists the reported results.

III. REPLICATION METHODOLOGY

A. Released Material

The authors' repository [9] contains the raw CSI dataset, the collection firmware, scripts for amplitude extraction, EMD and dataset construction, the two trained Keras models with their 8-bit TFLite conversions, and the array of real training samples. It contains no training script, no inference or preprocessing code for deployment, no ESP32-S3 firmware and no firmware build configuration.

B. Recovering the Split

The dataset-construction script holds out "the last six" subjects of an operating system directory listing, an order that is platform dependent. We matched each of the 1,198 samples in the released training array to a raw recording by correlation of the full 52×850 amplitude matrix. Every sample matched a distinct recording with 100% label agreement, and the six subjects absent from the array—s4 to s9—are the held-out set: 449 recordings, with per-class training and test counts identical to the paper's.

C. Recovering the Architecture

The released Keras model files embed their graph configuration. We rebuilt both models from it with identical layer names and verified the reconstruction by loading the authors' trained weights layer by layer and requiring numerically identical outputs. Two details differ from a conventional DenseNet: transition layers output a fixed five channels—the 0.5 compression factor is applied to the growth rate, $\lfloor 11 \times 0.5 \rfloor = 5$, not to the channel count—and transitions have no ReLU. The LSTM is unrolled, as TensorFlow Lite Micro requires. Both variants reproduce the reported parameter counts exactly.

D. Recovering the Input

Both released models take a $(850, 1)$ input, but neither the code nor the paper states how the single subcarrier is chosen. We probed the released small model with every one of the 52 subcarriers of every held-out recording (Section IV, Fig. 1); this determined both the training regime and the set of usable subcarriers.

E. Training

Following the probe, every usable subcarrier of every recording is a separate $(850, 1)$ sample. The augmentation reproduces the authors' scripts exactly (EEMD via PyEMD [13], 100 trials, noise seed 5, 7 modes plus residue, all within-subject, within-activity pairs), including their use of all 22 subjects; we call this the *paper protocol*. A *clean protocol* that augments only training subjects is implemented but not yet evaluated at full budget (Section VI).

EEMD of $1{,}647$ recordings $\times$ 47 subcarriers (77,409 series of 850 samples) took about 5 s per series, over a day of wall-clock time on a 10-core laptop CPU; EEMD is CPU-bound and does not benefit from a GPU. The resulting pool has 597,417 samples: 56,306 real and 541,111 synthetic. The synthetic count is exactly $11{,}513 \times 47$, matching the paper's 11,513 synthetic recordings.

Training used the published hyperparameters—Adam with learning rate 10^{-4}, batch 32, categorical cross-entropy, 300 epochs, each a full shuffled pass of 18,669 batches—on one NVIDIA T4 GPU (Kaggle, TensorFlow 2.20). Because sessions are limited to 12 h, training ran in three sessions resumed from per-epoch checkpoints that include optimizer state (Fig. 2), about 23 GPU-hours in total. No checkpoint selection was performed: the reported model is the one after epoch 300.

F. Quantization and Scoring

Models are converted with full-integer post-training quantization [14] (uint8 input and output, int8 internally), calibrated on 300 randomly drawn training-subject samples. We report three scorings of the held-out subjects:

- *per subcarrier*: each of the 47 subcarriers of each recording classified independently (21,103 decisions);
- *one subcarrier*: per recording, only the subcarrier with the largest variance of its min–max normalized amplitude among the 47—the on-device path, one inference per window;
- *vote*: majority vote over the 47 per-subcarrier decisions.

G. Hardware Reproduction

We built the collection system with two ESP32-D0WD-V3 development boards, one with a CP210x and one with a CH340 USB bridge. The released firmware ships without an ESP-IDF configuration; we built it with PlatformIO (ESP-IDF 5.5) and supplied the settings it cannot run without: CSI support enabled in the Wi-Fi driver (it is compiled out by default), the custom partition table the 1.4 MB application requires, and a 1 kHz FreeRTOS tick. We also patched the configuration-reset handler, which erased the stored configuration on any falling edge of GPIO0. On development boards GPIO0 is driven by the USB-serial auto-reset circuit, and in our tests opening the serial port was sufficient to erase the configuration; the patch requires the BOOT button to be held for 3 s. With the Rx streaming binary frames over USB, the host received 50.1–51.5 valid 128-byte frames per second.

The authors did not release their host capture script or inference firmware. Our host tool parses the binary stream and writes recordings in the dataset's format. The ESP32-S3 inference firmware windows incoming frames, selects the subcarrier and quantizes the input in plain C; this preprocessing is checked against a frozen output of the Python implementation on a real recording and agrees exactly in the selected subcarrier, all 850 values and every quantized byte.

IV. Findings About the Original Evaluation

F1 – Test subjects leak into training through augmentation: The authors' synthesis script pairs recordings within every subject directory, and the dataset-construction script places all synthetic samples in the training set. Recombined recordings of the six test subjects are therefore trained on. The counts confirm it: 22 subjects $\times$ 5 activities $\times\binom{15}{2} \approx 11{,}550$ against the reported 11,513, whereas the 16 training subjects alone would yield about 8,400. Our reproduction of the scripts yields exactly 11,513 synthetic recordings. Results under this protocol, including ours in Table II, measure accuracy on people whose data contributed to training.

F2 – The model is trained and evaluated per subcarrier: Although deployment uses a single subcarrier, training treats each subcarrier of each recording as an independent sample. The reported all-subcarrier latency of 10,904 ms is 47×232 ms: the single-subcarrier model run 47 times.

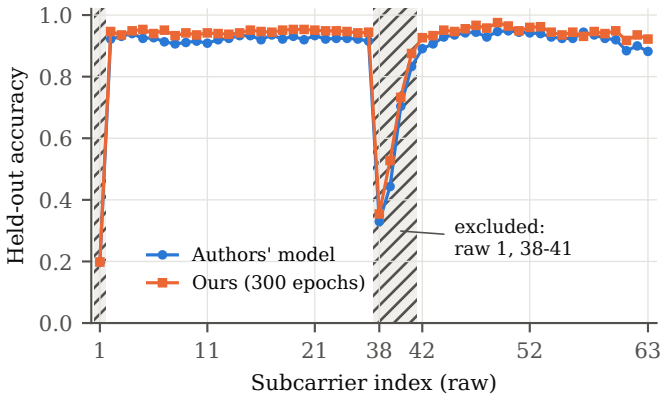


Fig. 1. Held-out accuracy of each kept subcarrier (paper protocol, float). Hatched bands mark the five subcarriers excluded to obtain the paper's 47.

F3 – The five "problematic" subcarriers: The paper removes five subcarriers without naming them. Fig. 1 shows that on raw subcarriers 1 and 38–41, adjacent to the DC null and the guard band, the released model scores 20–83%, against 88–95% on every other subcarrier (92–98% for ours). These subcarriers also have the highest normalized variance, because they are dominated by noise; a variance-based selection that does not exclude them selects the worst available input. In our first implementation it selected raw subcarrier 38 for a held-out recording.

F4 – What the headline numbers measure: Scored as released, the small model reaches 0.9243 per subcarrier in *floating point*—the value the paper reports as its 8-bit result—while the released 8-bit model scores 0.9030. The large model's floating-point majority vote, 0.9755, equals the paper's reported floating-point accuracy for that model (Table II).

F5 – Model size: The released 8-bit models are 325,376 and 779,680 bytes; our own conversion of the small model is 322,320 bytes. The reported 127 kB and 277 kB therefore do not describe the model files; they may refer to the tensor arena, which we have not yet measured on the device.

F6 – Released firmware is not runnable as shipped: As described in Section III, the collection firmware requires configuration that is not provided (CSI support, partition table) and loses its configuration on USB serial activity.

V. Results

Table II compares the released models and ours on the held-out subjects under the paper protocol. In floating point, our small model exceeds the released small model on all three scorings, by 2.1 points per subcarrier, 1.6 points for the single on-device subcarrier and 0.2 points for the vote. After 8-bit quantization it is 2.9 points ahead on the on-device path, and within 0.7 points per subcarrier and for the vote.

Per-class recall of our floating-point model per subcarrier is 0.96 (FA), 0.94 (GU), 0.95 (LD), 0.91 (SD) and 0.97 (WA). Quantization costs 4.7 points per subcarrier, consistent with the paper's reported loss of about 4%, and falls unevenly: sit-down recall drops from 0.91 to 0.76, mostly to get-up and fall. The 8-bit model is 322,320 bytes.

TABLE II
HELD-OUT ACCURACY, SUBJECTS S4–S9, PAPER PROTOCOL

Model	Prec.	Per subc.	One subc.	Vote
Released, small	float	0.9243	0.9198	0.9688
	uint8	0.9030	0.8797	0.9577
Released, large	float	0.9544	0.9532	0.9755
	uint8	0.9474	0.9376	0.9777
Ours, small	float	**0.9456**	**0.9354**	**0.9710**
	uint8	0.8982	**0.9087**	0.9510

Per subcarrier: 21,103 decisions; others: 449 recordings. Paper protocol: test subjects' recordings are in the synthetic training set (F1); not a subject-independent score.

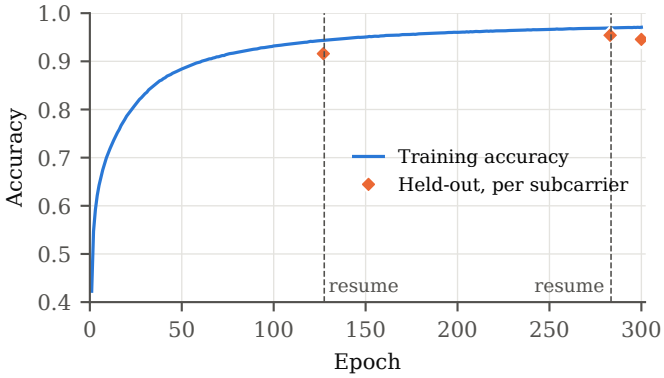


Fig. 2. Training accuracy per epoch over three resumed GPU sessions (dashed lines), with held-out per-subcarrier accuracy at the end of each session.

Fig. 2 shows training accuracy across the three resumed sessions and held-out scores at the session boundaries. Training accuracy rose smoothly to 0.971 without discontinuities at the resumption points.

A. Cross-Environment Test

On our hardware the host classifies a live 850-packet window every second in about 1.3 ms with the 8-bit model on a laptop CPU. To quantify transfer to a new environment, one author recorded the five activities and an additional empty-room class (EM) in a room on our campus, 10 repetitions of 20 s each, with the two boards of Section III on channel 1, the least congested 2.4 GHz channel at the site. Four of the 60 recordings were shorter than 850 packets because the stream stalled, leaving 56. Fine-tuning replaces the output layer with a six-class layer and trains with the paper's optimizer for 30 epochs; it is scored by stratified five-fold cross-validation over repetitions, so each recording is tested once by a model that did not see it. When EEMD augmentation is used, synthetic samples are built from each fold's training recordings only. Table III gives the results.

Unchanged, the paper-trained model is close to chance in the new room and labels the empty room mostly as get-up or lie-down. Neither fine-tuning nor fine-tuning with the paper's own augmentation recovers it. The fit check separates the possible explanations: initialized from the paper-trained weights, the network fits the recordings it is trained on almost perfectly, so the training procedure works and the recordings are mutually distinguishable, but what it learns does not transfer to held-out repetitions. From random initialization it cannot fit them at all, so the paper-trained representation is what makes the new data learnable.

TABLE III
NEW ROOM, ONE SUBJECT, 56 RECORDINGS

Setting	Cls.	Per subc.	One subc.	Vote
Unchanged	5	0.268	0.239	0.304
Fine-tuned	6	0.289	0.268	0.286
Fine-tuned + EEMD	6	0.241	0.196	0.268
Fine-tuned + gain norm.	6	0.337	0.286	0.393
From scratch	6	0.218	0.196	0.214
Fit, fine-tuned†	6	0.984	0.982	1.000
Fit, from scratch†	6	0.221	0.232	0.179

Chance 0.20 (5 classes), 0.167 (6). Upper rows: five-fold cross-validation.
†Trained and scored on all 56 recordings.

A model-free measurement explains why. We define motion energy as the mean absolute packet-to-packet change in amplitude, normalized by its mean, averaged over the 47 subcarriers. In the original dataset walking has the highest median motion energy (0.211, against 0.180–0.198 for the other activities). In our recordings the empty room has the highest (0.170, against 0.125–0.151). Dividing each packet's amplitudes by their mean over subcarriers—which cancels the receiver's automatic gain changes, since these scale all subcarriers of a packet together—removes about three quarters of the fluctuation (empty room 0.042, walking 0.039). Fine-tuning on gain-normalized amplitudes is the best setting we tested (Table III: 0.337 per subcarrier and 0.393 by vote, against a chance level of 0.167), but it remains far from usable, and in the motion-energy measure the activities still do not stand out from the empty room. Most of the variation in our recordings is therefore not caused by the subject, which indicates that link geometry and recording protocol, rather than the classifier, must be addressed first. Gain normalization is not part of the original preprocessing and is reported as a candidate addition to it.

VI. DISCUSSION AND LIMITATIONS

The reproduced numbers are not subject-independent. Because of F1, both the original results and our Table II measure performance on people whose recombined recordings were trained on. The clean protocol, which removes this leakage, is implemented but has only been run at a reduced CPU budget (8 augmented subcarriers, 150 epochs of 8,000 samples), where it reached 0.49 per-subcarrier accuracy while still improving; that run is under-trained and not comparable. A full-budget clean run is the most important next measurement.

Environment dependence. The original data come from one room, and our test in a second room (Section V-A) shows near-chance transfer. That test is itself limited: one subject, one session, 10 repetitions per class and a single link geometry. Its motion-energy analysis indicates that our recordings carried little subject-induced variation, so it does not yet show whether a better-placed link would transfer. In practical deployments the room is rarely empty; a single Tx–Rx link mixes the motion of everyone present, and separating

concurrent activities of several people is beyond the single-link design studied here.

Embedded measurements. We have not yet measured latency or tensor-arena size on an ESP32-S3, so the reported 232 ms and 127 kB remain unverified; the firmware is written and its preprocessing is host-verified, but has not been built and run on the device.

Scope of the replication. We trained the small model only, with one seed and one calibration set of 300 samples. We used the paper's 47 subcarriers; the released scripts retain 52, which would add the five noisy subcarriers to training.

VII. Future Work

1) Full-budget training and evaluation under the clean protocol.
2) On-device measurement of latency, arena size and accuracy on an ESP32-S3.
3) A second cross-environment collection with controlled link geometry, verified beforehand with the motion-energy check; 15 repetitions per class; and a "no event" class that includes background movement by other people, with each activity recorded both alone and with background activity. Recording volunteers other than the authors requires ethics approval.
4) Multiple seeds, the large model, a 52-subcarrier variant and a larger quantization calibration set.

VIII. Conclusion

We reproduced an embedded Wi-Fi sensing pipeline for human activity recognition from its public release and, under the original protocol, matched or exceeded the released model. The replication also showed that the original evaluation trains on augmented recordings of its test subjects, that the model is trained and scored per subcarrier, which five subcarriers are excluded and why, and that the headline 8-bit accuracy and model size do not correspond to the released artefacts. In a second room the model did not transfer, and neither fine-tuning nor the original augmentation recovered it; most of the signal variation there was unrelated to the subject. These details are necessary to interpret the published numbers and to compare future work fairly. We release the reconstructed pipeline so that the leak-free evaluation can be completed and extended.

Code and Data Availability

The code is available from the authors on request. It comprises preprocessing, split recovery, the probe of the released models, augmentation, training, quantization, evaluation, host capture, the build configuration and patch for the collection firmware, and the ESP32-S3 firmware with its host tests. The dataset and original models are those of [9].

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